Calculate gravitational force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate a gravitational interaction with consistent units.

SUCCESS CRITERIA - I can...

1. Identify both masses and the center separation.
2. Square the distance before dividing.
3. Check units and the scale of the result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “separation” and “mass” to explain your model or result. Sentence starters are optional.

separation: distance between the specified objects or centers

mass: a measure of how strongly an object resists acceleration

inverse square: proportional to one divided by distance squared

01 Bring back an earlier idea

A ball begins with downward velocity 3 m/s and falls 2 s.
Use downward positive, g=10 m/s² and no drag.
Find final velocity and displacement.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $F = Gm_1m_2/r^2$; $G = 6.67\times10^{-11}$ N·m²/kg².
r is center-to-center separation. CPA may use the supplied numerator.

Worked reasoning: In a supplied gravity model, $Gm_1m_2=18$ N·m² and r=3 m. Square first: $r^2=9$ m², so $F=18/9=2$ N, attractive.

Your turn: Sketch the two centers, label r and add the two force arrows.
Show how the units reduce to newtons.

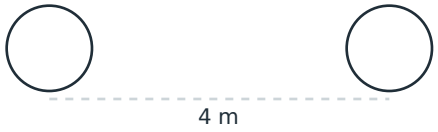
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

For spheres 4 m apart center to center, a student substitutes r=2 m because each is 2 m from the midpoint.
Identify the error.

Sentence starter: The step I would change is … . The evidence is … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Solve and justify

Two masses are 2.0×10^5 kg and 3.0×10^5 kg, separated by 10 m.
Calculate gravitational force. Then double separation without changing masses.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Analyze the reasoning

Why can a tiny force between classroom-sized objects be physically real even if you cannot feel it?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg box slides left at 6 m/s on a rough horizontal floor after release.
Draw forces and explain what follows about vertical and horizontal acceleration.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

Two equal 1000 kg masses are 2 m apart.
Find force. A student forgets to square 2: by what factor is their answer wrong?

Sentence starter: The step I would change is ... . The evidence is ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A pair has force 0.12 N at 2 m. Without finding either mass, calculate Gm_1m_2 and the separation for a force of 0.03 N.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3